

NGC 6569 CaT + SPACE/SP_Ace + MARCS SED Pipeline

Help File / User Manual for the v9 Jupyter Notebook

Notebook: NGC6569_CaT_SPACE_SED_full_pipeline_v9.ipynb

Prepared for Joanne Hughes - June 2026

Purpose: This manual explains how to run the v9 notebook from raw AAT FITS spectra through normalized `aat_*.txt` spectra, CaT radial velocities, Dias & Parisi CaT metallicities, SPACE/SP_Ace standard and alpha/metals fits, MARCS-cache SED fits, integrated diagnostic plots, and final reporting tables.

Reference diagnostic plot style

The v9 notebook should make integrated plots with the same scientific content as this fiber 215 comparison: a spectral truth model with residuals and a MARCS SED comparison panel produced with the original SEDfit_v2 logic.

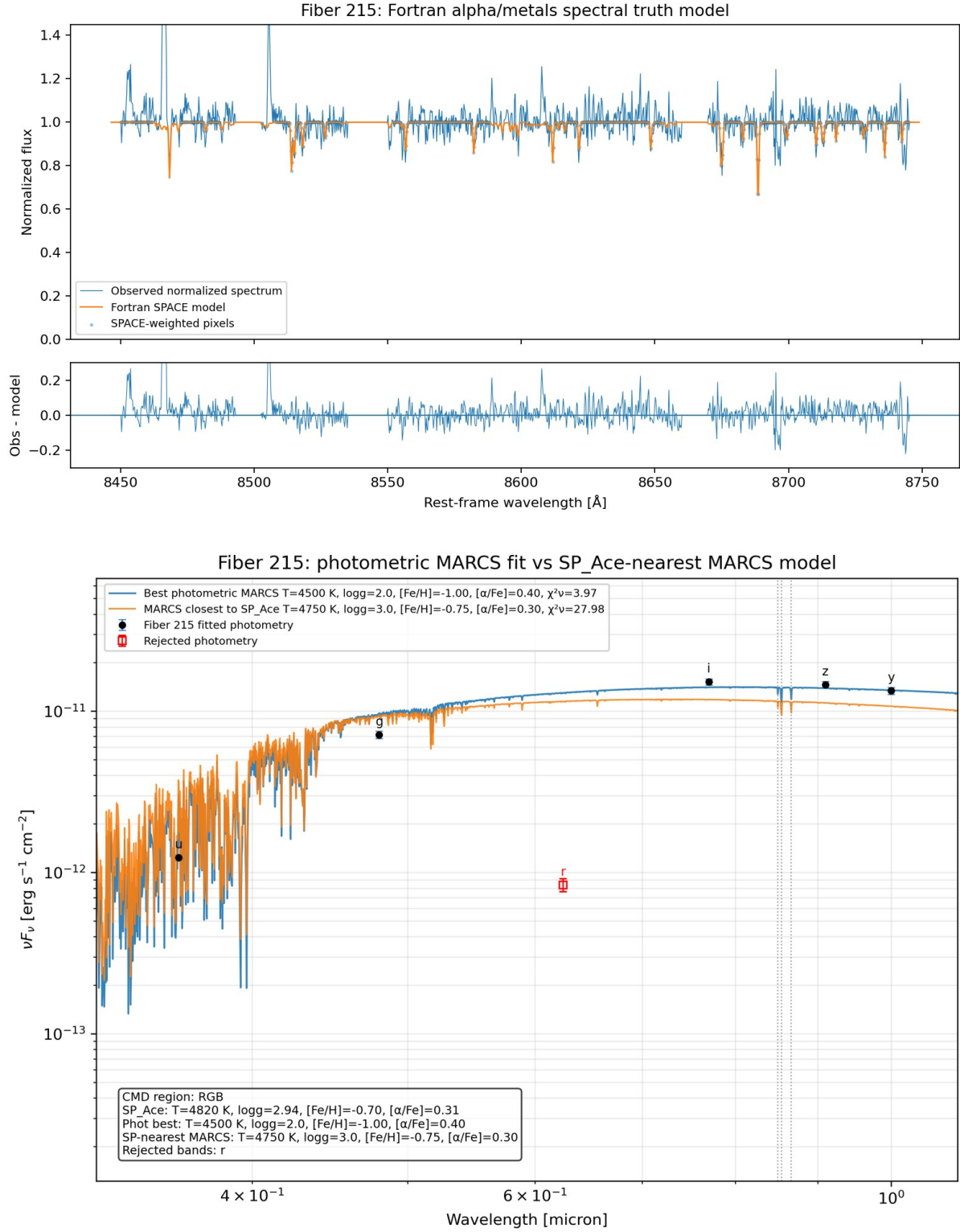


Figure 1. Expected integrated diagnostic format for one fiber: SPACE/SP_Ace spectral model and residuals above the SEDfit_v2 MARCS photometric comparison plot.

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1. What the v9 notebook does

The v9 notebook is a single integrated workflow for the NGC 6569 AAT/AAOmega Ca II triplet sample. It is designed to use all available information rather than trusting any one diagnostic alone. The intended final products are one row per fiber in a master table and one integrated diagnostic plot per fiber with valid spectral and SED information.

Diagnostic	Role in the pipeline	How to use it scientifically
CaT Voigt / XCSAO-style RV	Normalizes each spectrum, measures the CaT line positions and EWs, and shifts the spectrum to the stellar rest/lab frame.	Use RV_barycentric_report for kinematic membership. Inspect CaT line shape for bad sky subtraction, continuum errors, or poor line centering.
Dias & Parisi [Fe/H]DP	Uses CaT equivalent widths and the local BDBS i-band HB level to compute an independent CaT metallicity.	Use as an independent metallicity check and as fallback when SP_Ace is weak. Disagreement > about 0.30 dex is a caution flag.
SPACE/SP_Ace standard fit	Runs or ingests the full-line conservative no_norm fit.	Use for Teff,logg,[Fe/H] when the spectral model matches the broad absorption structure.
SPACE/SP_Ace alpha/metals fit	Runs or ingests the alpha/metals companion fit.	Use [alpha/Fe] only with caution, especially at low S/N. High S/N and a good model are required for strong claims.
MARCS SED fit	Uses dereddened BDBS ugrizy photometry and a cached MARCS grid, including leave-one-out band rejection and SP-nearest comparison.	Use as a physical plausibility and variability/photometry check. It is not a direct replacement for the spectrum.

Important scientific rule: A star is not accepted or rejected from one number alone. The best-practices result combines RV, CaT EW/[Fe/H]DP, SP_Ace/SPACE model quality, alpha/metals fit quality, MARCS SED plausibility, and visual inspection of the integrated plot.

2. Required input files and directories

The notebook is configured for Joanne Hughes' local Windows/WSL-style project layout. The paths can be changed in the USER CONFIGURATION cell. The most important correction is that the raw FITS files live in the NGC6569_spectra directory, not the SPACE directory.

Item	Default path in v9	Notes
Raw AAT FITS files	C:/Users/bella/Desktop/ NGC6569_spectra/FITS_FILES	Input FITS spectra. The notebook repairs bad RA/DEC header cards into a separate output directory and does not modify originals.
Main working directory	C:/Users/bella/Desktop/SPACE	Contains SPACE executable, GCOG library, tables, and MARCS cache.
Photometry/Gaia table	C:/Users/bella/Desktop/SPACE/ CaT_summary_BDBS_XCSAO.csv	Used for fiber IDs, BDBS magnitudes, Gaia data, and extinction-corrected photometry columns.
SP_Ace averages table	C:/Users/bella/Desktop/SPACE/ SP_Ace_averages_with_FeHDP_avg_std.csv	Used to merge existing SP_Ace values and uncertainties.
Best averages table	C:/Users/bella/Desktop/SPACE/ NGC6569_best_averages_alpha_rule_full.csv	Used when available as the most complete pre-existing adopted table.
MARCS cache	C:/Users/bella/Desktop/SPACE/ MARCS/ MARCS_synthetic_ugrizy_cache.csv	Precomputed MARCS ugrizy cache. This avoids rerunning the slow MARCS grid build.
GCOG library	C:/Users/bella/Desktop/SPACE/ GCOGlibrarySPACE1	Used by SPACE/SP_Ace. Preserve the working CaT-region library used for this project.
SPACE executable	C:/Users/bella/Desktop/SPACE/SPACE or SPACE.exe	Compiled Fortran executable. The notebook tries several common locations.

```
# Main configuration variables in v9
BASE_DIR = Path(r"C:/Users/bella/Desktop/SPACE")
RAW_FITS_DIR = Path(r"C:/Users/bella/Desktop/NGC6569_spectra/FITS_FILES")
OUT_DIR = BASE_DIR / "NGC6569_full_CaT_SPACE_SED_pipeline_v9"
PHOT_FILE = BASE_DIR / "CaT_summary_BDBS_XCSAO.csv"
SP_ACE_AVG_FILE = BASE_DIR / "SP_Ace_averages_with_FeHDP_avg_std.csv"
BEST_AVERAGES_FILE = BASE_DIR / "NGC6569_best_averages_alpha_rule_full.csv"
MARCS_CACHE_FILE = BASE_DIR / "MARCS" / "MARCS_synthetic_ugrizy_cache.csv"
GCOG_DIR = BASE_DIR / "GCOGlibrarySPACE1"
```

3. Quick start run procedure

Run the notebook from top to bottom, but test on a small number of fibers first. The test run should include at least one known good case, such as fiber 215, so that the integrated plot can be compared with the earlier reference plot.

1. Open NGC6569_CaT_SPACE_SED_full_pipeline_v9.ipynb in Jupyter.
2. In the USER CONFIGURATION cell, confirm BASE_DIR, RAW_FITS_DIR, PHOT_FILE, MARCS_CACHE_FILE, GCOG_DIR, and SPACE_EXECUTABLE.
3. Set MAX_FIBERS = 3 for a first test run. If you specifically want to test fiber 215, run the early stages and then use the display cell at the end with fiber_to_show = 215.

- Run the cells through Stage 5 and inspect the integrated plots. The SED plot should match the original SEDfit_v2 style, not the simplified incorrect redraw.
- When the test plots look correct, set MAX_FIBERS = None and rerun the full notebook.
- Check that the final compact table has one row per real NGC 6569 fiber. For the cleaned NGC 6569 sample this should be about 335 rows, not extra rows from BH261 or 11bm2 files.

```
# Test run
MAX_FIBERS = 3
RUN_STAGE0_FROM_FITS = True
RUN_SPACE_STANDARD = True
RUN_SPACE_ALPHA_METALS = True
RUN_SED_FITTING = True
MAKE_INTEGRATED_PLOTS = True

# Full run
MAX_FIBERS = None
```

4. Cell-by-cell guide

Notebook cell	Name	Main job
Cell 0	Markdown introduction	Explains the complete v9 workflow and why SEDfit_v2 logic is used for the MARCS plots.
Cell 1	USER CONFIGURATION	Set paths, run switches, max fibers, SPACE executable, output directories, and input tables.
Cell 2	GENERAL HELPERS	Defines fiber ID parsing, table readers, FITS discovery, two-column spectrum readers, and input table indexing.
Cell 3	FITS HEADER REPAIR + ROBUST FITS READER	Repairs bad AAT RA/DEC cards in copied FITS files and reads spectra robustly. Originals are preserved.
Cell 4	NORMALIZATION + CaT VOIGT FITS + BARYCENTRIC RV	Continuum normalizes spectra, fits CaT lines, estimates RV and S/N, writes lab-frame spectrum utilities.
Cell 5	STAGE 0 FITS -> aat_*.txt + CaT table	Reads repaired FITS, normalizes, fits CaT, shifts to rest/lab frame, writes aat_<fiber>.txt and CaT master table.
Cell 6	STAGE 1 Dias & Parisi [Fe/H]DP	Prompts for local imag_HB and computes reduced CaT EW and [Fe/H]DP.
Cell 7	STAGE 2 SPACE/SP_Ace runs	Writes space.par files and runs standard and alpha/metals SPACE fits, or ingests existing model files.
Cell 8-9	STAGE 3 MARCS SED fitting	Runs the original SEDfit_v2-style MARCS-cache fitter with leave-one-out rejected band logic and SP-nearest comparison.
Cell 10	STAGE 4 SPACE model + integrated diagnostics	Defines spectral model readers, sky/outlier masking, and the integrated spectral+SED plot function.

Cell 11	STAGE 5 integrated plots for all fibers	Generates one fiber_<N>_spectral_truth_plus_SED.png plot for each fiber with valid model and SED information.
Cell 12	STAGE 6 final summary tables	Merges CaT/DP, SP_Ace, SED, best averages, and plot paths into full and publication tables.
Cell 13	Display one complete comparison	Shows a selected fiber, typically 215, to verify the final diagnostic plot.

5. Detailed stage descriptions

5.1 Stage 0 - FITS to normalized rest-frame aat_*.txt

Stage 0 is the part that has to happen before the SP_Ace-like work. The raw AAT FITS files are read, repaired if needed, continuum normalized, fit for CaT line velocities, shifted into the stellar rest/lab frame, and written as aat_<fiber>.txt files. These are the spectra that SPACE/SP_Ace expects.

- Input: raw FITS files in RAW_FITS_DIR.
- Output spectra: OUT_DIR/aat_normalized_lab_frame/aat_<fiber>.txt.
- Output table: OUT_DIR/tables/CaT_master_from_FITS_normalized_labframe.csv.
- Frame convention: input is observed/topocentric; output wavelengths are stellar rest/lab-frame after applying RVraw_CaT.
- Barycentric RV is stored separately as RV_bary_CaT and is used for reporting and membership.

Header repair: The AAT FITS files can contain illegal unquoted RA/DEC cards. v9 makes repaired copies in OUT_DIR/repaired_fits_for_pipeline. Do not edit the original FITS files.

```
# Main Stage 0 output files
OUT_DIR / "aat_normalized_lab_frame" / "aat_<fiber>.txt"
OUT_DIR / "tables" / "CaT_master_from_FITS_normalized_labframe.csv"
OUT_DIR / "tables" / "fits_header_repair_manifest.csv"
```

5.2 Avoiding duplicate fibers from other fields

If the FITS directory contains spectra from other fields, such as BH261 or 11bm2, a generic parser can assign the same numerical fiber ID to several different stars. This creates duplicate final rows. For NGC 6569-only runs, the FITS finder should be restricted to n6569 files, or the final table should be checked for one row per fiber.

```
def find_fits_files(raw_dir):
    raw_dir = Path(raw_dir)
    patterns = [
        "n6569_2_*.fits", "n6569_2_*.fit", "n6569_2_*.fts",
        "N6569_2_*.fits", "N6569_2_*.fit", "N6569_2_*.fts",
    ]
    files = []
    for pat in patterns:
        files.extend(raw_dir.glob(pat))
    return sorted(set(files), key=lambda p: str(p).lower())

# Final duplicate guard after master is created
master = master.drop_duplicates().copy()
if "fiberid" in master.columns:
    dup_counts = master["fiberid"].value_counts()
```

```

dups = dup_counts[dup_counts > 1]
if len(dups) > 0:
    print("Duplicate fiber IDs remain:")
    print(dups)
    raise ValueError("Final table still has duplicate fiber IDs.")

```

5.3 Stage 1 - Dias & Parisi CaT metallicity

Stage 1 calculates an independent CaT-based metallicity. The user is prompted for the local horizontal branch/red clump reference level `imag_HB`. Use the apparent BDBS i-band HB level for the local field, not an extinction-corrected value, unless deliberately testing a different calibration.

Quantity	Meaning
<code>imag_HB</code>	Local apparent BDBS i-band magnitude of the HB/red clump used for the luminosity correction.
<code>SumEW23</code>	Sum of the two stronger CaT line equivalent widths, <code>EW8542</code> + <code>EW8662</code> , when using a two-line relation.
<code>EW3L</code> or <code>SumEW123</code>	Three-line equivalent-width sum after conversion or direct measurement.
<code>W'_i</code>	Reduced CaT equivalent width after the luminosity correction.
<code>[Fe/H]DP</code>	Dias & Parisi CaT metallicity estimate. Use as an independent check on <code>SP_Ace [Fe/H]</code> .

```

# Conceptual Dias & Parisi calculation used in the project
W_i_prime = SumEW3L + beta_i * (imag - imag_HB)
FeH_DP = -2.966 + 0.362 * W_i_prime

```

```

# Project caution: if SP_Ace [Fe/H] and [Fe/H]DP differ by > about 0.30 dex,
# the final reporting table should flag that star for caution or review.

```

5.4 Stage 2 - SPACE/SP_Ace standard and alpha/metals runs

Stage 2 writes a per-fiber `space.par` file and calls the compiled Fortran SPACE executable. It runs two modes: the standard full-line conservative `no_norm` fit and the alpha/metals companion fit. The notebook can also ingest existing model files if the run flags are set to False.

Mode	Main output	Scientific use
standard	<code>space_model_<fiber>_best_conservative_no_norm.dat</code> and TGM/ABD files	Best broad spectral fit for <code>Teff</code> , <code>logg</code> , <code>[M/H]</code> , <code>[Fe/H]</code> when reliable.
alpha_metals	<code>space_model_alpha_metals_<fiber>_best_conservative_no_norm.dat</code> and TGM/ABD files	Companion fit for alpha/metals. <code>[alpha/Fe]</code> is most useful at adequate S/N and with good model agreement.

```

# Conservative CaT-region wavelength windows used for SPACE/SP_Ace
wave_lims 8450 8493 8503 8535 8550 8660 8670 8745
no_norm
error_est
ABD_loop
Salaris_MH
# alpha is included only for the alpha/metals mode

```

5.5 Stage 3 - MARCS SED fitting

Stage 3 is intentionally based on the original `SEDfit_v2` leave-one-out MARCS-cache fitting cell. This is important because a simplified redraw of the SED plot gave the wrong visual behavior. The v9 notebook should reproduce the `SEDfit_v2` plot style: photometric points, best photometric MARCS model, `SP_Ace`-nearest MARCS model, rejected-band marking, and the lower-left information box.

- The photometry is dereddened by using $x_0 = x_{\text{mag}} - A_x$ for each BDBS filter $x = u, g, r, i, z, y$.
- The MARCS grid is read from a cached synthetic ugrizy file rather than rebuilt from scratch.
- The SED fit is a photometric consistency check, not a replacement for the spectrum.
- A rejected band or poor SP-nearest MARCS χ^2 should create a caution flag, especially for RGB/AGB/HB variability or non-simultaneous photometry.

Variability caution: RGB, AGB, and HB stars can vary. A mismatch between the spectrum and the photometric SED is not automatically a failed spectral fit. Treat it as a caution flag and inspect the diagnostic plot.

5.6 Stages 4-5 - integrated spectral + SED diagnostic plots

The integrated plot is the primary visual QC product. It combines the SPACE/SP_Ace truth model and residuals with the SEDfit_v2 MARCS photometric fit. These plots should be made for every fiber with a valid spectral model and SED fit.

Panel	What to inspect
Top spectral panel	Observed normalized spectrum versus SPACE/SP_Ace model. Ignore narrow sky-line residual spikes. Check broad absorption structure and continuum agreement.
Spectral residual panel	Residuals should be centered near zero over accepted windows. Narrow sky spikes are expected and should not dominate the decision.
SED panel	Best MARCS photometric model versus SP_Ace-nearest MARCS model; inspect rejected bands, UV/blue mismatch, and whether SP parameters are physically plausible.

```
# Integrated plot output
OUT_DIR / "integrated_CaT_SPACE_SED_diagnostics" / \
  "fiber_<fiber>_spectral_truth_plus_SED.png"
```

6. Output products

Output	Path relative to OUT_DIR	Description
Repaired FITS copies	repaired_fits_for_pipeline/	Copies of raw FITS files with bad RA/DEC cards neutralized. Original files are unchanged.
Normalized lab-frame spectra	aat_normalized_lab_frame/ aat_<fiber>.txt	Two-column wavelength [Å] and normalized flux files for SPACE/SP_Ace.
CaT master table	tables/ CaT_master_from_FITS_normalized_l abframe.csv	CaT RV, barycentric RV, EW, S/N, frame, and FITS provenance.
CaT + DP table	tables/ CaT_master_with_Dias_Parisi_FeHDP .csv	CaT master table plus Dias & Parisi [Fe/H]DP calculation.
SPACE run log	tables/SPACE_run_log.csv	Per-fiber run status for standard and alpha/metals SPACE runs.
SED summary table	tables/ MARCS_batch_restricted_fit_summar y_LOO_reject_SPclosest_MARCS.csv	MARCS-cache SED fit results with LOO rejection and SP-nearest model comparison.
Integrated plot manifest	tables/ integrated_diagnostic_plot_manifest. csv	One row per integrated PNG, including model and SED plot paths.

Full master table	tables/ NGC6569_full_integrated_best_practices_master_table.csv	Full merged table with all available diagnostics and reporting columns.
Publication reporting table	tables/ NGC6569_publication_best_practices_reporting_table.csv	Compact table for paper/report use. One row per fiber after duplicate checks.

7. How final reported values are chosen

The v9 notebook creates reporting columns so that the final table can preserve all methods while still giving one best-practices value per star. The goal is not to hide disagreements, but to make them explicit.

Reported quantity	Preferred source	Fallback / caution
RV_barycentric_report	CaT/XCSAO-style barycentric RV from the CaT analysis.	This is the membership RV. If unavailable, use existing RV columns but flag missing barycentric correction.
Teff_report	Adopted or SP_Ace Teff when the spectral model is acceptable.	SED Teff can be used as a plausibility check, not blindly substituted.
logg_report	Adopted or SP_Ace logg when the spectral model is acceptable.	SED logg disagreement can indicate variability, photometric mismatch, or SP fit tension.
FeH_report	Adopted/SP_Ace [Fe/H] when model and CaT-DP agree reasonably.	Use CaT-DP as fallback/check. Flag if $ \text{[Fe/H]}_{\text{SP}} - \text{[Fe/H]}_{\text{DP}} > 0.30$ dex.
alphaFe_report	Alpha/metals SP_Ace-like value for good spectra.	At low S/N or poor alpha/metals model quality, report as caution or omit from strong interpretation.

```
# Main final tables
full_out = TABLE_DIR / "NGC6569_full_integrated_best_practices_master_table.csv"
compact_out = TABLE_DIR / "NGC6569_publication_best_practices_reporting_table.csv"
```

8. Quality-control rules and cautions

- Inspect the integrated plot for every important candidate. The table flags are guides, not a substitute for visual inspection.
- Do not penalize a fit for narrow residual sky-line spikes, especially in the red. Judge the broad model agreement over the accepted SP_Ace windows.
- Use CaT RV as the primary velocity diagnostic for membership.
- Use [Fe/H]DP as an independent metallicity check. A disagreement larger than about 0.30 dex should be documented.
- Use SED mismatch as a caution flag, not an automatic rejection, because the photometry and spectroscopy are not necessarily simultaneous and RGB/AGB/HB stars can vary.
- Require extra caution for alpha abundance claims at low S/N. The red spectra are strongest for RV and metallicity, not always for detailed alpha abundances.
- Keep one row per fiber in final output. If duplicates appear, first check whether non-NGC 6569 FITS files were included in the raw directory.

9. Troubleshooting

Problem	Likely cause	Fix
KeyError: fiberid after Stage 0	No FITS files found, or all files failed before normal rows were made.	Check RAW_FITS_DIR, file names, and use the safer Stage 0D cell that guarantees a fiberid column.
Unparsable card (RA)	AAT FITS header has illegal RA or DEC card.	Use v9 FITS repair cell. It replaces bad RA/DEC cards in copied FITS files only.
SED plot looks wrong or nearly flat	The simplified integrated SED redraw was used instead of SEDfit_v2 logic.	Use v9 SEDfit_v2-based Stage 3. Do not use the earlier simplified SED plotting cell.
Some fibers have multiple entries	The FITS directory includes other fields such as BH261 or 11bm2, and the parser used only the number as fiberid.	Filter FITS files to n6569_2_*.fits or run the duplicate guard before writing final tables.
SPACE executable not found	SPACE or SPACE.exe is not in one of the candidate locations.	Set SPACE_EXECUTABLE explicitly in the configuration cell.
No integrated plots	Model files or SED PNGs were not produced, or MAX_FIBERS was too restrictive.	Check SPACE_run_log.csv, SED summary CSV, and integrated_diagnostic_plot_manifest.csv.
Too many failed SED fits	MARCS cache path is wrong or photometry/extinction columns are missing.	Verify MARCS_CACHE_FILE and that the photometry file contains xmag and Ax or equivalent columns.

10. Re-running or modifying the pipeline

Task	Switches / action
Rerun from raw FITS	RUN_STAGE0_FROM_FITS = True
Use existing normalized aat files	RUN_STAGE0_FROM_FITS = False; ensure aat_<fiber>.txt files already exist in NORMALIZED_AAT_DIR.
Rerun SPACE standard only	RUN_SPACE_STANDARD = True; RUN_SPACE_ALPHA_METALS = False
Rerun alpha/metals only	RUN_SPACE_STANDARD = False; RUN_SPACE_ALPHA_METALS = True
Skip SPACE and only ingest previous model files	Set both SPACE run switches False, but make sure find_model_file can locate the model files.
Rerun SED fitting	RUN_SED_FITTING = True and SKIP_EXISTING_SED_PNG = False if you want to overwrite plots.
Rebuild integrated plots only	Set earlier run switches False as desired, keep MAKE_INTEGRATED_PLOTS = True, rerun Stages 4-5.

```
# Show the complete comparison for one fiber
from IPython.display import Image, display
fiber_to_show = 215
png = INTEGRATED_PLOT_DIR / f"fiber_{fiber_to_show}_spectral_truth_plus_SED.png"
print(png)
print("Exists?", png.exists())
if png.exists():
    display(Image(filename=str(png)))
```

11. Final run checklist

1. Confirm RAW_FITS_DIR points to C:/Users/bella/Desktop/NGC6569_spectra/FITS_FILES.

2. Restrict the FITS finder to NGC 6569 files if other cluster files are in the same directory.
3. Run with MAX_FIBERS = 3 and inspect at least one known diagnostic, preferably fiber 215.
4. Confirm the SED panel matches the SEDfit_v2 plot style.
5. Confirm integrated plots appear in integrated_CaT_SPACE_SED_diagnostics/.
6. Set MAX_FIBERS = None and rerun the full sample.
7. Check that the final publication table has one row per NGC 6569 fiber.
8. Open the high-priority candidate plots and manually inspect any RV/FeH/SED caution cases.
9. Use the compact reporting table for paper tables and the full master table for machine-readable output.

Appendix A. Column interpretation for the final tables

Column or family	Meaning
fiberid	Unique AAT fiber/star identifier. Final table should have one row per fiber.
RVraw_CaT	Raw CaT RV measured from observed/topocentric spectrum.
RV_bary_CaT / RV_barycentric_report	Barycentric RV used for membership and reporting.
CaT1, CaT2, CaT3, SumEW23	CaT equivalent-width measurements.
FeH_DP	Dias & Parisi CaT metallicity. Independent check/fallback.
Teff_report, logg_report, FeH_report, alphaFe_report	Best-practices reported atmospheric/chemical quantities.
FeH_SP_vs_DP_delta	Difference between SP_Ace [Fe/H] and CaT-DP [Fe/H]. Large values create a caution flag.
FeH_DP_caution	True when SP and CaT-DP metallicities disagree beyond the project threshold.
SED_* columns	MARCS SED fit outputs, rejected bands, chi2 values, SP-nearest MARCS comparison, and caution diagnostics.
integrated_plot	Path to the combined spectral truth + SED diagnostic PNG.
recommended_reporting_note	Short text flag explaining how cautiously to report the star.